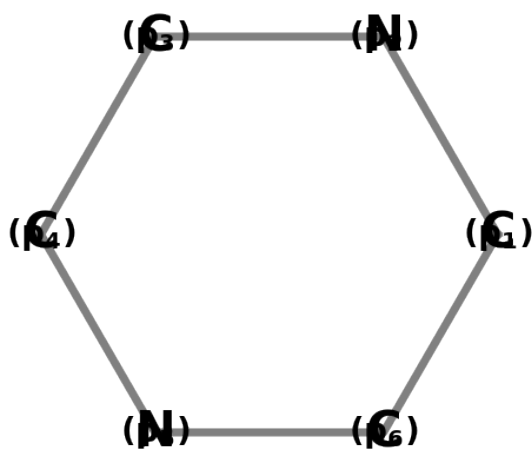


Pyrazine



Point Group: **D_{2h}**

p orbitals

Symmetry behavior of the p-orbitals:

$$\Gamma_{red} = 6E \oplus -2C2(y) \oplus -6\sigma(xy) \oplus 2\sigma(yz)$$

$$\Gamma_{irred} = 1B2g \oplus 2B3g \oplus 1Au \oplus 2B1u$$

- SALC of B3g: $\frac{\sqrt{2}(\mathbf{p}_1 - \mathbf{p}_4)}{2}$
- SALC of B1u: $\frac{\sqrt{2}(\mathbf{p}_1 + \mathbf{p}_4)}{2}$
- SALC of B2g: $\frac{\mathbf{p}_2}{2} + \frac{\mathbf{p}_3}{2} - \frac{\mathbf{p}_5}{2} - \frac{\mathbf{p}_6}{2}$
- SALC of B3g: $\frac{\mathbf{p}_2}{2} - \frac{\mathbf{p}_3}{2} - \frac{\mathbf{p}_5}{2} + \frac{\mathbf{p}_6}{2}$
- SALC of Au: $\frac{\mathbf{p}_2}{2} - \frac{\mathbf{p}_3}{2} + \frac{\mathbf{p}_5}{2} - \frac{\mathbf{p}_6}{2}$
- SALC of B1u: $\frac{\mathbf{p}_2}{2} + \frac{\mathbf{p}_3}{2} + \frac{\mathbf{p}_5}{2} + \frac{\mathbf{p}_6}{2}$

Put together these SALCs form the following Hamilton matrices:

$$H_{B3g} = \begin{bmatrix} \tilde{\alpha}_N & \sqrt{2}\tilde{\beta}_N \\ \sqrt{2}\tilde{\beta}_N & \tilde{\alpha} - \tilde{\beta} \end{bmatrix}$$

$$H_{B1u} = \begin{bmatrix} \tilde{\alpha}_N & \sqrt{2}\tilde{\beta}_N \\ \sqrt{2}\tilde{\beta}_N & \tilde{\alpha} + \tilde{\beta} \end{bmatrix}$$

$$H_{B2g} = [\tilde{\alpha} + \tilde{\beta}]$$

$$H_{Au} = [\tilde{\alpha} - \tilde{\beta}]$$

Solving Hückels secular equations, that follow from these H, leads to:

- orbital of B1u with energy = $\frac{\tilde{\alpha}}{2} + \frac{\tilde{\alpha}_N}{2} + \frac{\tilde{\beta}}{2} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2}$
(

$$\left[\sqrt{8 \left| \frac{\tilde{\beta}_N}{\tilde{\alpha} - \tilde{\alpha}_N + \tilde{\beta} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}} \right|^2} + 1 \left(\tilde{\alpha} - \tilde{\alpha}_N + \tilde{\beta} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2} \right) \sqrt{8 \left| \frac{\tilde{\beta}_N}{\tilde{\alpha} - \tilde{\alpha}_N + \tilde{\beta} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}} \right|^2} + 1 \left(\tilde{\alpha} - \tilde{\alpha}_N + \tilde{\beta} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2} \right) \right] \\ = \frac{4\tilde{\beta}_N (\mathbf{p}_1 + \mathbf{p}_4) + \left(\tilde{\alpha} - \tilde{\alpha}_N + \tilde{\beta} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2} \right) (\mathbf{p}_2 + \mathbf{p}_3)}{2\sqrt{8 \left| \frac{\tilde{\beta}_N}{\tilde{\alpha} - \tilde{\alpha}_N + \tilde{\beta} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}} \right|^2} + 1 \left(\tilde{\alpha} - \tilde{\alpha}_N + \tilde{\beta} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2} \right)}$$

- orbital of B1u with energy = $\frac{\tilde{\alpha}}{2} + \frac{\tilde{\alpha}_N}{2} + \frac{\tilde{\beta}}{2} + \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2}$
(

$$\left[\sqrt{8 \left| \frac{\tilde{\beta}_N}{\tilde{\alpha} - \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}} \right|^2} + 1 \left(\tilde{\alpha} - \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2} \right) \sqrt{8 \left| \frac{\tilde{\beta}_N}{\tilde{\alpha} - \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}} \right|^2} + 1 \left(\tilde{\alpha} - \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2} \right) \right]$$

s orbitals

Symmetry behavior of the s-orbitals:

$$\Gamma_{red} = 6E \oplus 2C2(y) \oplus 6\sigma(xy) \oplus 2\sigma(yz)$$

$$\Gamma_{irred} = 2Ag \oplus 1B1g \oplus 2B2u \oplus 1B3u$$

- SALC of Ag: $\frac{\sqrt{2}(s_1+s_4)}{2}$
- SALC of B2u: $\frac{\sqrt{2}(s_1-s_4)}{2}$
- SALC of Ag: $\frac{s_2}{2} + \frac{s_3}{2} + \frac{s_5}{2} + \frac{s_6}{2}$
- SALC of B1g: $\frac{s_2}{2} - \frac{s_3}{2} + \frac{s_5}{2} - \frac{s_6}{2}$
- SALC of B2u: $\frac{s_2}{2} - \frac{s_3}{2} - \frac{s_5}{2} + \frac{s_6}{2}$
- SALC of B3u: $\frac{s_2}{2} + \frac{s_3}{2} - \frac{s_5}{2} - \frac{s_6}{2}$

Put together these SALCs form the following Hamilton matrices:

$$H_{Ag} = \begin{bmatrix} \alpha_{s_N} & \sqrt{2}\beta_{s_N} \\ \sqrt{2}\beta_{s_N} & \tilde{\alpha}_s + \tilde{\beta}_s \end{bmatrix}$$

$$H_{B2u} = \begin{bmatrix} \alpha_{s_N} & \sqrt{2}\beta_{s_N} \\ \sqrt{2}\beta_{s_N} & \tilde{\alpha}_s - \tilde{\beta}_s \end{bmatrix}$$

$$H_{B1g} = [\tilde{\alpha}_s - \tilde{\beta}_s]$$

$$H_{B3u} = [\tilde{\alpha}_s + \tilde{\beta}_s]$$

Solving Hückels secular equations, that follow from these H, leads to:

- orbital of Ag with energy = $\frac{\tilde{\alpha}_s}{2} + \frac{\alpha_{s_N}}{2} + \frac{\tilde{\beta}_s}{2} - \frac{\sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_N} + 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_N}^2 - 2\alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2}}{2}$
(

$$\left[\sqrt{8 \left| \frac{\beta_{s_N}}{\tilde{\alpha}_s - \alpha_{s_N} + \tilde{\beta}_s - \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_N} + 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_N}^2 - 2\alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2}} \right|^2} + 1 \right] \frac{2\sqrt{2}\beta_{s_N}}{\tilde{\alpha}_s - \alpha_{s_N} + \tilde{\beta}_s - \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_N} + 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_N}^2 - 2\alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2}} + 1$$

$$= \frac{4\beta_{s_N}(s_1 + s_4) + \left(\tilde{\alpha}_s - \alpha_{s_N} + \tilde{\beta}_s - \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_N} + 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_N}^2 - 2\alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2} \right)}{2\sqrt{8 \left| \frac{\beta_{s_N}}{\tilde{\alpha}_s - \alpha_{s_N} + \tilde{\beta}_s - \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_N} + 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_N}^2 - 2\alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2}} \right|^2} + 1 \left(\tilde{\alpha}_s - \alpha_{s_N} + \tilde{\beta}_s - \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_N} + 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_N}^2 - 2\alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2} \right)}$$

- orbital of Ag with energy = $\frac{\tilde{\alpha}_s}{2} + \frac{\alpha_{s_N}}{2} + \frac{\tilde{\beta}_s}{2} + \frac{\sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_N} + 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_N}^2 - 2\alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2}}{2}$
(

$$\left[\sqrt{8 \left| \frac{\beta_{s_N}}{\tilde{\alpha}_s - \alpha_{s_N} + \tilde{\beta}_s + \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_N} + 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_N}^2 - 2\alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2}} \right|^2} + 1 \right] \frac{2\sqrt{2}\beta_{s_N}}{\tilde{\alpha}_s - \alpha_{s_N} + \tilde{\beta}_s + \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_N} + 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_N}^2 - 2\alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2}} + 1$$

Transitions from p orbitals into s orbitals

State with Occupation [2, 2, 2, 0, 0, 0] (0 unpaired electrons, mult 1.0)

Transition from ψ_2 into ψ_{s_2} :

- $-[2, 2, 2, 0, 0, 0], (0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 2, 0, 0, 0), (0, 1, 0, 0, 0, 0)$
- ψ_2 : linear combination =
$$\frac{4\tilde{\beta}_N(\mathbf{p}_1 - \mathbf{p}_4)}{\sqrt{8 \left| \frac{\tilde{\beta}_N}{-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}} \right|^2 + 1} \left(\tilde{\alpha} - \tilde{\alpha}_N - \tilde{\beta} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2} \right)}$$

energy =
$$\frac{\tilde{\alpha}}{2} + \frac{\tilde{\alpha}_N}{2} - \frac{\tilde{\beta}}{2} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2}$$
- η_2 : linear combination =
$$\frac{4\beta_{s_N}(\mathbf{s}_1 - \mathbf{s}_4)}{\sqrt{8 \left| \frac{\beta_{s_N}}{-\tilde{\alpha}_s + \alpha_{s_N} + \tilde{\beta}_s + \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_N} - 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_N}^2 + 2\alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2}} \right|^2 + 1} \left(\tilde{\alpha}_s - \alpha_{s_N} - \tilde{\beta}_s - \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_N} - 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_N}^2 + 2\alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2} \right)}$$

energy =
$$\frac{\tilde{\alpha}_s}{2} + \frac{\alpha_{s_N}}{2} - \frac{\tilde{\beta}_s}{2} - \frac{\sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_N} - 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_N}^2 + 2\alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2}}{2}$$
- Δ Energy:
$$-\frac{\tilde{\alpha}}{2} - \frac{\tilde{\alpha}_N}{2} + \frac{\tilde{\alpha}_s}{2} + \frac{\alpha_{s_N}}{2} + \frac{\tilde{\beta}}{2} - \frac{\tilde{\beta}_s}{2} + \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2} - \frac{\sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_N} - 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_N}^2 + 2\alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2}}{2}$$
- Energy of State before Excitation:
$$4\tilde{\alpha} + 2\tilde{\alpha}_N + 2\tilde{\beta} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}$$
- Energy of State after Excitation:
$$\frac{7\tilde{\alpha}}{2} + \frac{3\tilde{\alpha}_N}{2} + \frac{\tilde{\alpha}_s}{2} + \frac{\alpha_{s_N}}{2} + \frac{5\tilde{\beta}}{2} - \frac{\tilde{\beta}_s}{2} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2} - \frac{\sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_N} - 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_N}^2 + 2\alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2}}{2}$$
- Energy between Average Energies of ψ -/ η -orbitals:
$$-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_N}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{\alpha_{s_N}}{3}$$
- dipole transition moment:
$$\frac{\tilde{\delta} \left(8\tilde{\beta}_N\beta_{s_N} + \left(-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2} \right) \right)}{\sqrt{8 \left| \frac{\tilde{\beta}_N}{-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}} \right|^2 + 1} \sqrt{8 \left| \frac{\beta_{s_N}}{-\tilde{\alpha}_s + \alpha_{s_N} + \tilde{\beta}_s + \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_N} - 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_N}^2 + 2\alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2}} \right|^2 + 1}}$$

Transition from ψ_3 into ψ_{s_3} :

- $-[2, 2, 2, 0, 0, 0], (0, 0, 0, 0, 0, 0) \rightarrow (2, 2, 1, 0, 0, 0), (0, 0, 1, 0, 0, 0)$
- ψ_3 : linear combination = $\frac{\mathbf{p}_2}{2} + \frac{\mathbf{p}_3}{2} - \frac{\mathbf{p}_5}{2} - \frac{\mathbf{p}_6}{2}$, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_3 : linear combination = $\frac{\mathbf{s}_2}{2} + \frac{\mathbf{s}_3}{2} - \frac{\mathbf{s}_5}{2} - \frac{\mathbf{s}_6}{2}$, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy:
$$-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$$
- Energy of State before Excitation:
$$4\tilde{\alpha} + 2\tilde{\alpha}_N + 2\tilde{\beta} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}$$
- Energy of State after Excitation:
$$3\tilde{\alpha} + 2\tilde{\alpha}_N + \tilde{\alpha}_s + \tilde{\beta} + \tilde{\beta}_s - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}$$
- Energy between Average Energies of ψ -/ η -orbitals:
$$-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_N}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{\alpha_{s_N}}{3}$$

- dipole transition moment: $\tilde{\delta}$

- AND

State with Occupation (2, 1, 1, 1, 1, 0) (4 unpaired electrons, mult 5.0)

Transition from ψ_2 into ψ_{s_2} :

- $-(2, 1, 1, 1, 1, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 0, 1, 1, 1, 0), (0, 1, 0, 0, 0, 0)$
- ψ_2 : linear combination = $\frac{4\tilde{\beta}_N(\mathbf{p}_1 - \mathbf{p}_4)}{\sqrt{8 \left| \frac{\tilde{\beta}_N}{-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}} \right|^2 + 1} \left(\tilde{\alpha} - \tilde{\alpha}_N - \tilde{\beta} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2} \right)}$
 energy = $\frac{\tilde{\alpha}}{2} + \frac{\tilde{\alpha}_N}{2} - \frac{\tilde{\beta}}{2} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2}$
- η_2 : linear combination = $\frac{4\beta_{s_N}(\mathbf{s}_1 - \mathbf{s}_4)}{\sqrt{8 \left| \frac{\beta_{s_N}}{-\tilde{\alpha}_s + \alpha_{s_N} + \tilde{\beta}_s + \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_N} - 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_N}^2 + 2\alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2}} \right|^2 + 1} \left(\tilde{\alpha}_s - \alpha_{s_N} - \tilde{\beta}_s - \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_N} - 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_N}^2 + 2\alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2} \right)}$
 energy = $\frac{\tilde{\alpha}_s}{2} + \frac{\alpha_{s_N}}{2} - \frac{\tilde{\beta}_s}{2} - \frac{\sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_N} - 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_N}^2 + 2\alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2}}{2}$
- Δ Energy: $-\frac{\tilde{\alpha}}{2} - \frac{\tilde{\alpha}_N}{2} + \frac{\tilde{\alpha}_s}{2} + \frac{\alpha_{s_N}}{2} + \frac{\tilde{\beta}}{2} - \frac{\tilde{\beta}_s}{2} + \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2} - \frac{\sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_N} - 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_N}^2 + 2\alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2}}{2}$
- Energy of State before Excitation: $4\tilde{\alpha} + 2\tilde{\alpha}_N + \tilde{\beta} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2}$
- Energy of State after Excitation: $\frac{7\tilde{\alpha}}{2} + \frac{3\tilde{\alpha}_N}{2} + \frac{\tilde{\alpha}_s}{2} + \frac{\alpha_{s_N}}{2} + \frac{3\tilde{\beta}}{2} - \frac{\tilde{\beta}_s}{2} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2} - \frac{\sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_N} - 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_N}^2 + 2\alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2}}{2}$
- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_N}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{\alpha_{s_N}}{3}$
- dipole transition moment: $\frac{\tilde{\delta} \left(8\tilde{\beta}_N\beta_{s_N} + \left(-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2} \right) \left(-\tilde{\alpha}_s + \alpha_{s_N} + \tilde{\beta}_s + \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_N} - 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_N}^2 + 2\alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2} \right) \right)}{\sqrt{8 \left| \frac{\tilde{\beta}_N}{-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}} \right|^2 + 1} \sqrt{8 \left| \frac{\beta_{s_N}}{-\tilde{\alpha}_s + \alpha_{s_N} + \tilde{\beta}_s + \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_N} - 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_N}^2 + 2\alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2}} \right|^2 + 1}}$

Transition from ψ_3 into ψ_{s_3} :

- $-(2, 1, 1, 1, 1, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 0, 1, 1, 0), (0, 0, 1, 0, 0, 0)$
- ψ_3 : linear combination = $\frac{\mathbf{p}_2}{2} + \frac{\mathbf{p}_3}{2} - \frac{\mathbf{p}_5}{2} - \frac{\mathbf{p}_6}{2}$, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_3 : linear combination = $\frac{\mathbf{s}_2}{2} + \frac{\mathbf{s}_3}{2} - \frac{\mathbf{s}_5}{2} - \frac{\mathbf{s}_6}{2}$, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$
- Energy of State before Excitation: $4\tilde{\alpha} + 2\tilde{\alpha}_N + \tilde{\beta} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2}$
- Energy of State after Excitation: $3\tilde{\alpha} + 2\tilde{\alpha}_N + \tilde{\alpha}_s + \tilde{\beta}_s - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2}$

- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_N}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{\alpha_{s_N}}{3}$
- dipole transition moment: $\tilde{\delta}$

- AND

Transition from ψ_4 into ψ_{s_4} :

- $-(2, 1, 1, 1, 1, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 1, 0, 1, 0), (0, 0, 0, 1, 0, 0)$

- ψ_4 : linear combination = $\frac{4\tilde{\beta}_N(\mathbf{p}_1+\mathbf{p}_4)}{\sqrt{8\left|\frac{\tilde{\beta}_N}{\tilde{\alpha}-\tilde{\alpha}_N+\tilde{\beta}+\sqrt{\tilde{\alpha}^2-2\tilde{\alpha}\tilde{\alpha}_N+2\tilde{\alpha}\tilde{\beta}+\tilde{\alpha}_N^2-2\tilde{\alpha}_N\tilde{\beta}+\tilde{\beta}^2+8\tilde{\beta}_N^2}}\right|^2+1}\left(\tilde{\alpha}-\tilde{\alpha}_N+\tilde{\beta}+\sqrt{\tilde{\alpha}^2-2\tilde{\alpha}\tilde{\alpha}_N+2\tilde{\alpha}\tilde{\beta}+\tilde{\alpha}_N^2-2\tilde{\alpha}_N\tilde{\beta}+\tilde{\beta}^2+8\tilde{\beta}_N^2}\right)}$
 energy = $\frac{\tilde{\alpha}}{2} + \frac{\tilde{\alpha}_N}{2} + \frac{\tilde{\beta}}{2} + \frac{\sqrt{\tilde{\alpha}^2-2\tilde{\alpha}\tilde{\alpha}_N+2\tilde{\alpha}\tilde{\beta}+\tilde{\alpha}_N^2-2\tilde{\alpha}_N\tilde{\beta}+\tilde{\beta}^2+8\tilde{\beta}_N^2}}{2}$

- η_4 : linear combination = $\frac{4\beta_{s_N}(\mathbf{s}_1+\mathbf{s}_4)}{\sqrt{8\left|\frac{\beta_{s_N}}{\tilde{\alpha}_s-\alpha_{s_N}+\tilde{\beta}_s+\sqrt{\tilde{\alpha}_s^2-2\tilde{\alpha}_s\alpha_{s_N}+2\tilde{\alpha}_s\tilde{\beta}_s+\alpha_{s_N}^2-2\alpha_{s_N}\tilde{\beta}_s+\tilde{\beta}_s^2+8\beta_{s_N}^2}}\right|^2+1}\left(\tilde{\alpha}_s-\alpha_{s_N}+\tilde{\beta}_s+\sqrt{\tilde{\alpha}_s^2-2\tilde{\alpha}_s\alpha_{s_N}+2\tilde{\alpha}_s\tilde{\beta}_s+\alpha_{s_N}^2-2\alpha_{s_N}\tilde{\beta}_s+\tilde{\beta}_s^2+8\beta_{s_N}^2}\right)}$
 energy = $\frac{\tilde{\alpha}_s}{2} + \frac{\alpha_{s_N}}{2} + \frac{\tilde{\beta}_s}{2} + \frac{\sqrt{\tilde{\alpha}_s^2-2\tilde{\alpha}_s\alpha_{s_N}+2\tilde{\alpha}_s\tilde{\beta}_s+\alpha_{s_N}^2-2\alpha_{s_N}\tilde{\beta}_s+\tilde{\beta}_s^2+8\beta_{s_N}^2}}{2}$

- Δ Energy: $-\frac{\tilde{\alpha}}{2} - \frac{\tilde{\alpha}_N}{2} + \frac{\tilde{\alpha}_s}{2} + \frac{\alpha_{s_N}}{2} - \frac{\tilde{\beta}}{2} + \frac{\tilde{\beta}_s}{2} - \frac{\sqrt{\tilde{\alpha}^2-2\tilde{\alpha}\tilde{\alpha}_N+2\tilde{\alpha}\tilde{\beta}+\tilde{\alpha}_N^2-2\tilde{\alpha}_N\tilde{\beta}+\tilde{\beta}^2+8\tilde{\beta}_N^2}}{2} + \frac{\sqrt{\tilde{\alpha}_s^2-2\tilde{\alpha}_s\alpha_{s_N}+2\tilde{\alpha}_s\tilde{\beta}_s+\alpha_{s_N}^2-2\alpha_{s_N}\tilde{\beta}_s+\tilde{\beta}_s^2+8\beta_{s_N}^2}}{2}$

- Energy of State before Excitation: $4\tilde{\alpha} + 2\tilde{\alpha}_N + \tilde{\beta} - \frac{\sqrt{\tilde{\alpha}^2-2\tilde{\alpha}\tilde{\alpha}_N+2\tilde{\alpha}\tilde{\beta}+\tilde{\alpha}_N^2-2\tilde{\alpha}_N\tilde{\beta}+\tilde{\beta}^2+8\tilde{\beta}_N^2}}{2} - \frac{\sqrt{\tilde{\alpha}^2-2\tilde{\alpha}\tilde{\alpha}_N+2\tilde{\alpha}\tilde{\beta}+\tilde{\alpha}_N^2-2\tilde{\alpha}_N\tilde{\beta}+\tilde{\beta}^2+8\tilde{\beta}_N^2}}{2}$

- Energy of State after Excitation: $\frac{7\tilde{\alpha}}{2} + \frac{3\tilde{\alpha}_N}{2} + \frac{\tilde{\alpha}_s}{2} + \frac{\alpha_{s_N}}{2} + \frac{\tilde{\beta}}{2} + \frac{\tilde{\beta}_s}{2} - \frac{\sqrt{\tilde{\alpha}^2-2\tilde{\alpha}\tilde{\alpha}_N+2\tilde{\alpha}\tilde{\beta}+\tilde{\alpha}_N^2-2\tilde{\alpha}_N\tilde{\beta}+\tilde{\beta}^2+8\tilde{\beta}_N^2}}{2} - \frac{\sqrt{\tilde{\alpha}^2-2\tilde{\alpha}\tilde{\alpha}_N+2\tilde{\alpha}\tilde{\beta}+\tilde{\alpha}_N^2-2\tilde{\alpha}_N\tilde{\beta}+\tilde{\beta}^2+8\tilde{\beta}_N^2}}{2} + \frac{\sqrt{\tilde{\alpha}_s^2-2\tilde{\alpha}_s\alpha_{s_N}+2\tilde{\alpha}_s\tilde{\beta}_s+\alpha_{s_N}^2-2\alpha_{s_N}\tilde{\beta}_s+\tilde{\beta}_s^2+8\beta_{s_N}^2}}{2}$

- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_N}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{\alpha_{s_N}}{3}$

- dipole transition moment: $\frac{\tilde{\delta}(8\tilde{\beta}_N\beta_{s_N}+(\tilde{\alpha}-\tilde{\alpha}_N+\tilde{\beta}+\sqrt{\tilde{\alpha}^2-2\tilde{\alpha}\tilde{\alpha}_N+2\tilde{\alpha}\tilde{\beta}+\tilde{\alpha}_N^2-2\tilde{\alpha}_N\tilde{\beta}+\tilde{\beta}^2+8\tilde{\beta}_N^2}))}{\sqrt{8\left|\frac{\tilde{\beta}_N}{\tilde{\alpha}-\tilde{\alpha}_N+\tilde{\beta}+\sqrt{\tilde{\alpha}^2-2\tilde{\alpha}\tilde{\alpha}_N+2\tilde{\alpha}\tilde{\beta}+\tilde{\alpha}_N^2-2\tilde{\alpha}_N\tilde{\beta}+\tilde{\beta}^2+8\tilde{\beta}_N^2}}\right|^2+1}\sqrt{8\left|\frac{\beta_{s_N}}{\tilde{\alpha}_s-\alpha_{s_N}+\tilde{\beta}_s+\sqrt{\tilde{\alpha}_s^2-2\tilde{\alpha}_s\alpha_{s_N}+2\tilde{\alpha}_s\tilde{\beta}_s+\alpha_{s_N}^2-2\alpha_{s_N}\tilde{\beta}_s+\tilde{\beta}_s^2+8\beta_{s_N}^2}}\right|^2+1}}$

- AND

Transition from ψ_5 into ψ_{s_5} :

- $-(2, 1, 1, 1, 1, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 1, 1, 0, 0), (0, 0, 0, 0, 1, 0)$

- ψ_5 : linear combination = $\frac{\mathbf{p}_2}{2} - \frac{\mathbf{p}_3}{2} + \frac{\mathbf{p}_5}{2} - \frac{\mathbf{p}_6}{2}$, energy = $\tilde{\alpha} - \tilde{\beta}$

- η_5 : linear combination = $\frac{\mathbf{s}_2}{2} - \frac{\mathbf{s}_3}{2} + \frac{\mathbf{s}_5}{2} - \frac{\mathbf{s}_6}{2}$, energy = $\tilde{\alpha}_s - \tilde{\beta}_s$

- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s$

- Energy of State before Excitation: $4\tilde{\alpha} + 2\tilde{\alpha}_N + \tilde{\beta} - \frac{\sqrt{\tilde{\alpha}^2-2\tilde{\alpha}\tilde{\alpha}_N+2\tilde{\alpha}\tilde{\beta}+\tilde{\alpha}_N^2-2\tilde{\alpha}_N\tilde{\beta}+\tilde{\beta}^2+8\tilde{\beta}_N^2}}{2} - \frac{\sqrt{\tilde{\alpha}^2-2\tilde{\alpha}\tilde{\alpha}_N+2\tilde{\alpha}\tilde{\beta}+\tilde{\alpha}_N^2-2\tilde{\alpha}_N\tilde{\beta}+\tilde{\beta}^2+8\tilde{\beta}_N^2}}{2}$

- Energy of State after Excitation: $3\tilde{\alpha} + 2\tilde{\alpha}_N + \tilde{\alpha}_s + 2\tilde{\beta} - \tilde{\beta}_s - \frac{\sqrt{\tilde{\alpha}^2-2\tilde{\alpha}\tilde{\alpha}_N+2\tilde{\alpha}\tilde{\beta}+\tilde{\alpha}_N^2-2\tilde{\alpha}_N\tilde{\beta}+\tilde{\beta}^2+8\tilde{\beta}_N^2}}{2} - \frac{\sqrt{\tilde{\alpha}^2-2\tilde{\alpha}\tilde{\alpha}_N+2\tilde{\alpha}\tilde{\beta}+\tilde{\alpha}_N^2-2\tilde{\alpha}_N\tilde{\beta}+\tilde{\beta}^2+8\tilde{\beta}_N^2}}{2}$

- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_N}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{\alpha_{s_N}}{3}$
- dipole transition moment: $\tilde{\delta}$

- AND

State with Occupation (2, 1, 2, 0, 1, 0) (2 unpaired electrons, mult 3.0)

Transition from ψ_2 into ψ_{s_2} :

- $-(2, 1, 2, 0, 1, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 0, 2, 0, 1, 0), (0, 1, 0, 0, 0, 0)$
- ψ_2 : linear combination = $\frac{4\tilde{\beta}_N(\mathbf{p}_1 - \mathbf{p}_4)}{\sqrt{8 \left| \frac{\tilde{\beta}_N}{-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}} \right|^2 + 1} \left(\tilde{\alpha} - \tilde{\alpha}_N - \tilde{\beta} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2} \right)}$
 energy = $\frac{\tilde{\alpha}}{2} + \frac{\tilde{\alpha}_N}{2} - \frac{\tilde{\beta}}{2} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2}$
- η_2 : linear combination = $\frac{4\beta_{s_N}(\mathbf{s}_1 - \mathbf{s}_4)}{\sqrt{8 \left| \frac{\beta_{s_N}}{-\tilde{\alpha}_s + \alpha_{s_N} + \tilde{\beta}_s + \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_N} - 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_N}^2 + 2\alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2}} \right|^2 + 1} \left(\tilde{\alpha}_s - \alpha_{s_N} - \tilde{\beta}_s - \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_N} - 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_N}^2 + 2\alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2} \right)}$
 energy = $\frac{\tilde{\alpha}_s}{2} + \frac{\alpha_{s_N}}{2} - \frac{\tilde{\beta}_s}{2} - \frac{\sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_N} - 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_N}^2 + 2\alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2}}{2}$
- Δ Energy: $-\frac{\tilde{\alpha}}{2} - \frac{\tilde{\alpha}_N}{2} + \frac{\tilde{\alpha}_s}{2} + \frac{\alpha_{s_N}}{2} + \frac{\tilde{\beta}}{2} - \frac{\tilde{\beta}_s}{2} + \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2} - \frac{\sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_N} - 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_N}^2 + 2\alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2}}{2}$
- Energy of State before Excitation: $\frac{9\tilde{\alpha}}{2} + \frac{3\tilde{\alpha}_N}{2} + \frac{3\tilde{\beta}}{2} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}$
- Energy of State after Excitation: $4\tilde{\alpha} + \tilde{\alpha}_N + \frac{\tilde{\alpha}_s}{2} + \frac{\alpha_{s_N}}{2} + 2\tilde{\beta} - \frac{\tilde{\beta}_s}{2} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2} - \frac{\sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_N} - 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_N}^2 + 2\alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2}}{2}$
- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_N}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{\alpha_{s_N}}{3}$
- dipole transition moment: $\frac{\tilde{\delta} \left(8\tilde{\beta}_N\beta_{s_N} + \left(-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2} \right) \left(-\tilde{\alpha}_s + \alpha_{s_N} + \tilde{\beta}_s + \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_N} - 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_N}^2 + 2\alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2} \right) \right)}{\sqrt{8 \left| \frac{\tilde{\beta}_N}{-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}} \right|^2 + 1} \sqrt{8 \left| \frac{\beta_{s_N}}{-\tilde{\alpha}_s + \alpha_{s_N} + \tilde{\beta}_s + \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_N} - 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_N}^2 + 2\alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2}} \right|^2 + 1}}$

Transition from ψ_3 into ψ_{s_3} :

- $-(2, 1, 2, 0, 1, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 1, 0, 1, 0), (0, 0, 1, 0, 0, 0)$
- ψ_3 : linear combination = $\frac{\mathbf{p}_2}{2} + \frac{\mathbf{p}_3}{2} - \frac{\mathbf{p}_5}{2} - \frac{\mathbf{p}_6}{2}$, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_3 : linear combination = $\frac{\mathbf{s}_2}{2} + \frac{\mathbf{s}_3}{2} - \frac{\mathbf{s}_5}{2} - \frac{\mathbf{s}_6}{2}$, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$
- Energy of State before Excitation: $\frac{9\tilde{\alpha}}{2} + \frac{3\tilde{\alpha}_N}{2} + \frac{3\tilde{\beta}}{2} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}$

- Energy of State after Excitation: $\frac{7\tilde{\alpha}}{2} + \frac{3\tilde{\alpha}_N}{2} + \tilde{\alpha}_s + \frac{\tilde{\beta}}{2} + \tilde{\beta}_s - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}$
- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_N}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{alpha_{s_N}}{3}$
- dipole transition moment: $\tilde{\delta}$

- AND

Transition from ψ_5 into ψ_{s_5} :

- $-(2, 1, 2, 0, 1, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 2, 0, 0, 0), (0, 0, 0, 0, 1, 0)$
- ψ_5 : linear combination = $\frac{p_2}{2} - \frac{p_3}{2} + \frac{p_5}{2} - \frac{p_6}{2}$, energy = $\tilde{\alpha} - \tilde{\beta}$
- η_5 : linear combination = $\frac{s_2}{2} - \frac{s_3}{2} + \frac{s_5}{2} - \frac{s_6}{2}$, energy = $\tilde{\alpha}_s - \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s$
- Energy of State before Excitation: $\frac{9\tilde{\alpha}}{2} + \frac{3\tilde{\alpha}_N}{2} + \frac{3\tilde{\beta}}{2} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}$
- Energy of State after Excitation: $\frac{7\tilde{\alpha}}{2} + \frac{3\tilde{\alpha}_N}{2} + \tilde{\alpha}_s + \frac{5\tilde{\beta}}{2} - \tilde{\beta}_s - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}$
- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_N}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{alpha_{s_N}}{3}$
- dipole transition moment: $\tilde{\delta}$

- AND

State with Occupation (2, 1, 2, 1, 0, 0) (2 unpaired electrons, mult 3.0)

Transition from ψ_2 into ψ_{s_2} :

- $-(2, 1, 2, 1, 0, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 0, 2, 1, 0, 0), (0, 1, 0, 0, 0, 0)$
- ψ_2 : linear combination = $\frac{4\tilde{\beta}_N(p_1 - p_4)}{\sqrt{8 \left| \frac{\tilde{\beta}_N}{-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}} \right|^2 + 1} (\tilde{\alpha} - \tilde{\alpha}_N - \tilde{\beta} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2})}$
energy = $\frac{\tilde{\alpha}}{2} + \frac{\tilde{\alpha}_N}{2} - \frac{\tilde{\beta}}{2} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2}$
- η_2 : linear combination = $\frac{4\beta_{s_N}(s_1 - s_4)}{\sqrt{8 \left| \frac{\beta_{s_N}}{-\tilde{\alpha}_s + alpha_{s_N} + \tilde{\beta}_s + \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s alpha_{s_N} - 2\tilde{\alpha}_s\tilde{\beta}_s + alpha_{s_N}^2 + 2alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2}} \right|^2 + 1} (\tilde{\alpha}_s - alpha_{s_N} - \tilde{\beta}_s - \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s alpha_{s_N} - 2\tilde{\alpha}_s\tilde{\beta}_s + alpha_{s_N}^2 + 2alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2})}$
energy = $\frac{\tilde{\alpha}_s}{2} + \frac{alpha_{s_N}}{2} - \frac{\tilde{\beta}_s}{2} - \frac{\sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s alpha_{s_N} - 2\tilde{\alpha}_s\tilde{\beta}_s + alpha_{s_N}^2 + 2alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2}}{2}$
- Δ Energy: $-\frac{\tilde{\alpha}}{2} - \frac{\tilde{\alpha}_N}{2} + \frac{\tilde{\alpha}_s}{2} + \frac{alpha_{s_N}}{2} + \frac{\tilde{\beta}}{2} - \frac{\tilde{\beta}_s}{2} + \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2} - \frac{\sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s alpha_{s_N} - 2\tilde{\alpha}_s\tilde{\beta}_s + alpha_{s_N}^2 + 2alpha_{s_N}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{s_N}^2}}{2}$
- Energy of State before Excitation: $4\tilde{\alpha} + 2\tilde{\alpha}_N + 3\tilde{\beta} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}$

- Energy of State after Excitation: $\frac{7\tilde{\alpha}}{2} + \frac{3\tilde{\alpha}_N}{2} + \frac{\tilde{\alpha}_s}{2} + \frac{\alpha_{sN}}{2} + \frac{7\tilde{\beta}}{2} - \frac{\tilde{\beta}_s}{2} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2} - \frac{\sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{sN} - 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{sN}^2 + 2\alpha_{sN}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{sN}^2}}{2}$
- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_N}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{\alpha_{sN}}{3}$
- dipole transition moment: $\tilde{\delta} \left(8\tilde{\beta}_N\beta_{sN} + (-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}) \right)$

Transition from ψ_3 into ψ_{s3} :

- $-(2, 1, 2, 1, 0, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 1, 1, 0, 0), (0, 0, 1, 0, 0, 0)$
- ψ_3 : linear combination = $\frac{p_2}{2} + \frac{p_3}{2} - \frac{p_5}{2} - \frac{p_6}{2}$, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_3 : linear combination = $\frac{s_2}{2} + \frac{s_3}{2} - \frac{s_5}{2} - \frac{s_6}{2}$, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$
- Energy of State before Excitation: $4\tilde{\alpha} + 2\tilde{\alpha}_N + 3\tilde{\beta} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2}$
- Energy of State after Excitation: $3\tilde{\alpha} + 2\tilde{\alpha}_N + \tilde{\alpha}_s + 2\tilde{\beta} + \tilde{\beta}_s - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2}$
- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_N}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{\alpha_{sN}}{3}$
- dipole transition moment: $\tilde{\delta}$

- AND

Transition from ψ_4 into ψ_{s4} :

- $-(2, 1, 2, 1, 0, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 2, 0, 0, 0), (0, 0, 0, 1, 0, 0)$
- ψ_4 : linear combination = $\frac{4\tilde{\beta}_N(p_1 + p_4)}{\sqrt{8 \left| \frac{\tilde{\beta}_N}{\tilde{\alpha} - \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}} \right|^2 + 1 \left(\tilde{\alpha} - \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2} \right)}}$
energy = $\frac{\tilde{\alpha}}{2} + \frac{\tilde{\alpha}_N}{2} + \frac{\tilde{\beta}}{2} + \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2}$
- η_4 : linear combination = $\frac{4\beta_{sN}(s_1 + s_4)}{\sqrt{8 \left| \frac{\beta_{sN}}{\tilde{\alpha}_s - \alpha_{sN} + \tilde{\beta}_s + \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{sN} + 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{sN}^2 - 2\alpha_{sN}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{sN}^2}} \right|^2 + 1 \left(\tilde{\alpha}_s - \alpha_{sN} + \tilde{\beta}_s + \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{sN} + 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{sN}^2 - 2\alpha_{sN}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{sN}^2} \right)}}$
energy = $\frac{\tilde{\alpha}_s}{2} + \frac{\alpha_{sN}}{2} + \frac{\tilde{\beta}_s}{2} + \frac{\sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{sN} + 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{sN}^2 - 2\alpha_{sN}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{sN}^2}}{2}$
- Δ Energy: $-\frac{\tilde{\alpha}}{2} - \frac{\tilde{\alpha}_N}{2} + \frac{\tilde{\alpha}_s}{2} + \frac{\alpha_{sN}}{2} - \frac{\tilde{\beta}}{2} + \frac{\tilde{\beta}_s}{2} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2} + \frac{\sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{sN} + 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{sN}^2 - 2\alpha_{sN}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{sN}^2}}{2}$
- Energy of State before Excitation: $4\tilde{\alpha} + 2\tilde{\alpha}_N + 3\tilde{\beta} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2}$

- Energy of State after Excitation: $\frac{7\tilde{\alpha}}{2} + \frac{3\tilde{\alpha}_N}{2} + \frac{\tilde{\alpha}_s}{2} + \frac{\alpha_{sN}}{2} + \frac{5\tilde{\beta}}{2} + \frac{\tilde{\beta}_s}{2} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2} + \frac{\sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{sN} + 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{sN}^2 - 2\alpha_{sN}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{sN}^2}}{2}$
- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_N}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{\alpha_{sN}}{3}$
- dipole transition moment: $\frac{\tilde{\delta}(8\tilde{\beta}_N\beta_{sN} + (\tilde{\alpha} - \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}))}{\sqrt{8\left|\frac{\tilde{\beta}_N}{-\tilde{\alpha} - \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}\right|^2 + 1}\sqrt{8\left|\frac{\beta_{sN}}{-\tilde{\alpha}_s - \alpha_{sN} + \tilde{\beta}_s + \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{sN} + 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{sN}^2 - 2\alpha_{sN}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{sN}^2}}\right|^2 + 1}}$

- AND

State with Occupation (2, 2, 1, 0, 1, 0) (2 unpaired electrons, mult 3.0)

Transition from ψ_2 into ψ_{s_2} :

- $-(2, 2, 1, 0, 1, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 1, 0, 1, 0), (0, 1, 0, 0, 0, 0)$
- ψ_2 : linear combination = $\frac{4\tilde{\beta}_N(\mathbf{p}_1 - \mathbf{p}_4)}{\sqrt{8\left|\frac{\tilde{\beta}_N}{-\tilde{\alpha} - \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}\right|^2 + 1}\left(\tilde{\alpha} - \tilde{\alpha}_N - \tilde{\beta} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}\right) + 1\left(\tilde{\alpha}_s - \alpha_{sN} - \tilde{\beta}_s - \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{sN} + 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{sN}^2 - 2\alpha_{sN}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{sN}^2}\right)}}$
energy = $\frac{\tilde{\alpha}}{2} + \frac{\tilde{\alpha}_N}{2} - \frac{\tilde{\beta}}{2} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2}$
- η_2 : linear combination = $\frac{4\beta_{sN}(\mathbf{s}_1 - \mathbf{s}_4)}{\sqrt{8\left|\frac{\beta_{sN}}{-\tilde{\alpha}_s + \alpha_{sN} + \tilde{\beta}_s + \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{sN} + 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{sN}^2 - 2\alpha_{sN}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{sN}^2}}\right|^2 + 1}\left(\tilde{\alpha}_s - \alpha_{sN} - \tilde{\beta}_s - \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{sN} + 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{sN}^2 - 2\alpha_{sN}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{sN}^2}\right)}}$
energy = $\frac{\tilde{\alpha}_s}{2} + \frac{\alpha_{sN}}{2} - \frac{\tilde{\beta}_s}{2} - \frac{\sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{sN} + 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{sN}^2 - 2\alpha_{sN}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{sN}^2}}{2}$
- Δ Energy: $-\frac{\tilde{\alpha}}{2} - \frac{\tilde{\alpha}_N}{2} + \frac{\tilde{\alpha}_s}{2} + \frac{\alpha_{sN}}{2} + \frac{\tilde{\beta}}{2} - \frac{\tilde{\beta}_s}{2} + \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2} - \frac{\sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{sN} + 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{sN}^2 - 2\alpha_{sN}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{sN}^2}}{2}$
- Energy of State before Excitation: $4\tilde{\alpha} + 2\tilde{\alpha}_N - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}$
- Energy of State after Excitation: $\frac{7\tilde{\alpha}}{2} + \frac{3\tilde{\alpha}_N}{2} + \frac{\tilde{\alpha}_s}{2} + \frac{\alpha_{sN}}{2} + \frac{\tilde{\beta}}{2} - \frac{\tilde{\beta}_s}{2} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2} - \frac{\sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{sN} + 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{sN}^2 - 2\alpha_{sN}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{sN}^2}}{2}$
- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_N}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{\alpha_{sN}}{3}$
- dipole transition moment: $\frac{\tilde{\delta}(8\tilde{\beta}_N\beta_{sN} + (-\tilde{\alpha} - \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}))}{\sqrt{8\left|\frac{\tilde{\beta}_N}{-\tilde{\alpha} - \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}\right|^2 + 1}\sqrt{8\left|\frac{\beta_{sN}}{-\tilde{\alpha}_s + \alpha_{sN} + \tilde{\beta}_s + \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{sN} + 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{sN}^2 - 2\alpha_{sN}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{sN}^2}}\right|^2 + 1}}$

Transition from ψ_3 into ψ_{s_3} :

- $-(2, 2, 1, 0, 1, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 2, 0, 0, 1, 0), (0, 0, 1, 0, 0, 0)$
- ψ_3 : linear combination = $\frac{\mathbf{p}_2}{2} + \frac{\mathbf{p}_3}{2} - \frac{\mathbf{p}_5}{2} - \frac{\mathbf{p}_6}{2}$, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_3 : linear combination = $\frac{\mathbf{s}_2}{2} + \frac{\mathbf{s}_3}{2} - \frac{\mathbf{s}_5}{2} - \frac{\mathbf{s}_6}{2}$, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$

- Energy of State before Excitation: $4\tilde{\alpha}+2\tilde{\alpha}_N-\sqrt{\tilde{\alpha}^2-2\tilde{\alpha}\tilde{\alpha}_N-2\tilde{\alpha}\tilde{\beta}+\tilde{\alpha}_N^2+2\tilde{\alpha}_N\tilde{\beta}+\tilde{\beta}^2+8\tilde{\beta}_N^2}-\sqrt{\tilde{\alpha}^2-2\tilde{\alpha}\tilde{\alpha}_N+2\tilde{\alpha}\tilde{\beta}+\tilde{\alpha}_N^2-2\tilde{\alpha}_N\tilde{\beta}+\tilde{\beta}^2+8\tilde{\beta}_N^2}$
- Energy of State after Excitation: $3\tilde{\alpha}+2\tilde{\alpha}_N+\tilde{\alpha}_s-\tilde{\beta}+\tilde{\beta}_s-\sqrt{\tilde{\alpha}^2-2\tilde{\alpha}\tilde{\alpha}_N-2\tilde{\alpha}\tilde{\beta}+\tilde{\alpha}_N^2+2\tilde{\alpha}_N\tilde{\beta}+\tilde{\beta}^2+8\tilde{\beta}_N^2}-\sqrt{\tilde{\alpha}^2-2\tilde{\alpha}\tilde{\alpha}_N+2\tilde{\alpha}\tilde{\beta}+\tilde{\alpha}_N^2-2\tilde{\alpha}_N\tilde{\beta}+\tilde{\beta}^2+8\tilde{\beta}_N^2}$
- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{2\tilde{\alpha}}{3}-\frac{\tilde{\alpha}_N}{3}+\frac{2\tilde{\alpha}_s}{3}+\frac{\alpha_{sN}}{3}$
- dipole transition moment: $\tilde{\delta}$

- AND

Transition from ψ_5 into ψ_{s_5} :

- $-(2, 2, 1, 0, 1, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 2, 1, 0, 0, 0), (0, 0, 0, 0, 1, 0)$
- ψ_5 : linear combination = $\frac{p_2}{2} - \frac{p_3}{2} + \frac{p_5}{2} - \frac{p_6}{2}$, energy = $\tilde{\alpha} - \tilde{\beta}$
- η_5 : linear combination = $\frac{s_2}{2} - \frac{s_3}{2} + \frac{s_5}{2} - \frac{s_6}{2}$, energy = $\tilde{\alpha}_s - \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s$
- Energy of State before Excitation: $4\tilde{\alpha}+2\tilde{\alpha}_N-\sqrt{\tilde{\alpha}^2-2\tilde{\alpha}\tilde{\alpha}_N-2\tilde{\alpha}\tilde{\beta}+\tilde{\alpha}_N^2+2\tilde{\alpha}_N\tilde{\beta}+\tilde{\beta}^2+8\tilde{\beta}_N^2}-\sqrt{\tilde{\alpha}^2-2\tilde{\alpha}\tilde{\alpha}_N+2\tilde{\alpha}\tilde{\beta}+\tilde{\alpha}_N^2-2\tilde{\alpha}_N\tilde{\beta}+\tilde{\beta}^2+8\tilde{\beta}_N^2}$
- Energy of State after Excitation: $3\tilde{\alpha}+2\tilde{\alpha}_N+\tilde{\alpha}_s+\tilde{\beta}-\tilde{\beta}_s-\sqrt{\tilde{\alpha}^2-2\tilde{\alpha}\tilde{\alpha}_N-2\tilde{\alpha}\tilde{\beta}+\tilde{\alpha}_N^2+2\tilde{\alpha}_N\tilde{\beta}+\tilde{\beta}^2+8\tilde{\beta}_N^2}-\sqrt{\tilde{\alpha}^2-2\tilde{\alpha}\tilde{\alpha}_N+2\tilde{\alpha}\tilde{\beta}+\tilde{\alpha}_N^2-2\tilde{\alpha}_N\tilde{\beta}+\tilde{\beta}^2+8\tilde{\beta}_N^2}$
- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{2\tilde{\alpha}}{3}-\frac{\tilde{\alpha}_N}{3}+\frac{2\tilde{\alpha}_s}{3}+\frac{\alpha_{sN}}{3}$
- dipole transition moment: $\tilde{\delta}$

- AND

State with Occupation (2, 2, 1, 1, 0, 0) (2 unpaired electrons, mult 3.0)

Transition from ψ_2 into ψ_{s_2} :

- $-(2, 2, 1, 1, 0, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 1, 1, 0, 0), (0, 1, 0, 0, 0, 0)$
- ψ_2 : linear combination = $\frac{4\tilde{\beta}_N(p_1-p_4)}{\sqrt{8\left|\frac{\tilde{\beta}_N}{-\tilde{\alpha}+\tilde{\alpha}_N+\tilde{\beta}+\sqrt{\tilde{\alpha}^2-2\tilde{\alpha}\tilde{\alpha}_N-2\tilde{\alpha}\tilde{\beta}+\tilde{\alpha}_N^2+2\tilde{\alpha}_N\tilde{\beta}+\tilde{\beta}^2+8\tilde{\beta}_N^2}}\right|^2+1\left(\tilde{\alpha}-\tilde{\alpha}_N-\tilde{\beta}-\sqrt{\tilde{\alpha}^2-2\tilde{\alpha}\tilde{\alpha}_N-2\tilde{\alpha}\tilde{\beta}+\tilde{\alpha}_N^2+2\tilde{\alpha}_N\tilde{\beta}+\tilde{\beta}^2+8\tilde{\beta}_N^2}\right)^2}}$
energy = $\frac{\tilde{\alpha}}{2} + \frac{\tilde{\alpha}_N}{2} - \frac{\tilde{\beta}}{2} - \frac{\sqrt{\tilde{\alpha}^2-2\tilde{\alpha}\tilde{\alpha}_N-2\tilde{\alpha}\tilde{\beta}+\tilde{\alpha}_N^2+2\tilde{\alpha}_N\tilde{\beta}+\tilde{\beta}^2+8\tilde{\beta}_N^2}}{2}$
- η_2 : linear combination = $\frac{4\beta_{sN}(s_1-s_4)}{\sqrt{8\left|\frac{\beta_{sN}}{-\tilde{\alpha}_s+\alpha_{sN}+\tilde{\beta}_s+\sqrt{\tilde{\alpha}_s^2-2\tilde{\alpha}_s\alpha_{sN}-2\tilde{\alpha}_s\tilde{\beta}_s+\alpha_{sN}^2+2\alpha_{sN}\tilde{\beta}_s+\tilde{\beta}_s^2+8\beta_{sN}^2}}\right|^2+1\left(\tilde{\alpha}_s-\alpha_{sN}-\tilde{\beta}_s-\sqrt{\tilde{\alpha}_s^2-2\tilde{\alpha}_s\alpha_{sN}-2\tilde{\alpha}_s\tilde{\beta}_s+\alpha_{sN}^2+2\alpha_{sN}\tilde{\beta}_s+\tilde{\beta}_s^2+8\beta_{sN}^2}\right)^2}}$
energy = $\frac{\tilde{\alpha}_s}{2} + \frac{\alpha_{sN}}{2} - \frac{\tilde{\beta}_s}{2} - \frac{\sqrt{\tilde{\alpha}_s^2-2\tilde{\alpha}_s\alpha_{sN}-2\tilde{\alpha}_s\tilde{\beta}_s+\alpha_{sN}^2+2\alpha_{sN}\tilde{\beta}_s+\tilde{\beta}_s^2+8\beta_{sN}^2}}{2}$

- Δ Energy: $-\frac{\tilde{\alpha}}{2} - \frac{\tilde{\alpha}_N}{2} + \frac{\tilde{\alpha}_s}{2} + \frac{\alpha_{sN}}{2} + \frac{\tilde{\beta}}{2} - \frac{\tilde{\beta}_s}{2} + \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2} - \frac{\sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{sN} - 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{sN}^2}}{2}$
- Energy of State before Excitation: $\frac{7\tilde{\alpha}}{2} + \frac{5\tilde{\alpha}_N}{2} + \frac{3\tilde{\beta}}{2} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2}$
- Energy of State after Excitation: $3\tilde{\alpha} + 2\tilde{\alpha}_N + \frac{\tilde{\alpha}_s}{2} + \frac{\alpha_{sN}}{2} + 2\tilde{\beta} - \frac{\tilde{\beta}_s}{2} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2} - \frac{\sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{sN} - 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{sN}^2 + 2\alpha_{sN}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{sN}^2}}{2}$
- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_N}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{\alpha_{sN}}{3}$
- dipole transition moment: $\frac{\tilde{\delta}(8\tilde{\beta}_N\beta_{sN} + (-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}))}{\sqrt{8\left|\frac{\tilde{\beta}_N}{-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}\right|^2 + 1}\sqrt{8\left|\frac{\beta_{sN}}{-\tilde{\alpha}_s + \alpha_{sN} + \tilde{\beta}_s + \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{sN} - 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{sN}^2 + 2\alpha_{sN}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{sN}^2}}\right|^2 + 1}}$

Transition from ψ_3 into ψ_{s_3} :

- $-(2, 2, 1, 1, 0, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 2, 0, 1, 0, 0), (0, 0, 1, 0, 0, 0)$
- ψ_3 : linear combination = $\frac{p_2}{2} + \frac{p_3}{2} - \frac{p_5}{2} - \frac{p_6}{2}$, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_3 : linear combination = $\frac{s_2}{2} + \frac{s_3}{2} - \frac{s_5}{2} - \frac{s_6}{2}$, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$
- Energy of State before Excitation: $\frac{7\tilde{\alpha}}{2} + \frac{5\tilde{\alpha}_N}{2} + \frac{3\tilde{\beta}}{2} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2}$
- Energy of State after Excitation: $\frac{5\tilde{\alpha}}{2} + \frac{5\tilde{\alpha}_N}{2} + \tilde{\alpha}_s + \frac{\tilde{\beta}}{2} + \tilde{\beta}_s - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2}$
- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_N}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{\alpha_{sN}}{3}$
- dipole transition moment: $\tilde{\delta}$

- AND

Transition from ψ_4 into ψ_{s_4} :

- $-(2, 2, 1, 1, 0, 0), (0, 0, 0, 0, 0, 0) \rightarrow (2, 2, 1, 0, 0, 0), (0, 0, 0, 1, 0, 0)$
- ψ_4 : linear combination = $\frac{4\tilde{\beta}_N(p_1 + p_4)}{\sqrt{8\left|\frac{\tilde{\beta}_N}{\tilde{\alpha} - \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}\right|^2 + 1}(\tilde{\alpha} - \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2})}$
energy = $\frac{\tilde{\alpha}}{2} + \frac{\tilde{\alpha}_N}{2} + \frac{\tilde{\beta}}{2} + \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2}$
- η_4 : linear combination = $\frac{4\beta_{sN}(s_1 + s_4)}{\sqrt{8\left|\frac{\beta_{sN}}{\tilde{\alpha}_s - \alpha_{sN} + \tilde{\beta}_s + \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{sN} + 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{sN}^2 - 2\alpha_{sN}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{sN}^2}}\right|^2 + 1}(\tilde{\alpha}_s - \alpha_{sN} + \tilde{\beta}_s + \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{sN} + 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{sN}^2 - 2\alpha_{sN}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{sN}^2})}$
energy = $\frac{\tilde{\alpha}_s}{2} + \frac{\alpha_{sN}}{2} + \frac{\tilde{\beta}_s}{2} + \frac{\sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{sN} + 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{sN}^2 - 2\alpha_{sN}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{sN}^2}}{2}$
- Δ Energy: $-\frac{\tilde{\alpha}}{2} - \frac{\tilde{\alpha}_N}{2} + \frac{\tilde{\alpha}_s}{2} + \frac{\alpha_{sN}}{2} - \frac{\tilde{\beta}}{2} + \frac{\tilde{\beta}_s}{2} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2} + \frac{\sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{sN} + 2\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{sN}^2 - 2\alpha_{sN}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8\beta_{sN}^2}}{2}$

- Energy of State before Excitation: $\frac{7\tilde{\alpha}}{2} + \frac{5\tilde{\alpha}_N}{2} + \frac{3\tilde{\beta}}{2} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}}{2}$
- Energy of State after Excitation: $3\tilde{\alpha} + 2\tilde{\alpha}_N + \frac{\tilde{\alpha}_s}{2} + \frac{alpha_{sN}}{2} + \tilde{\beta} + \frac{\tilde{\beta}_s}{2} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2} + \frac{\sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s alpha_{sN} + 2\tilde{\alpha}_s\tilde{\beta}_s + alpha_{sN}^2 - 2alpha_{sN}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8beta_{sN}^2}}{2}$
- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_N}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{alpha_{sN}}{3}$
- dipole transition moment: $\frac{\tilde{\delta} \left(8\tilde{\beta}_N beta_{sN} + \left(\tilde{\alpha} - \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2} \right) \right)}{\sqrt{8 \left| \frac{\tilde{\beta}_N}{\tilde{\alpha} - \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N + 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 - 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}} \right|^2 + 1} \sqrt{8 \left| \frac{beta_{sN}}{\tilde{\alpha}_s - alpha_{sN} + \tilde{\beta}_s + \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s alpha_{sN} + 2\tilde{\alpha}_s\tilde{\beta}_s + alpha_{sN}^2 - 2alpha_{sN}\tilde{\beta}_s + \tilde{\beta}_s^2 + 8beta_{sN}^2}} \right|^2 + 1}}$

- AND

Dispersion energies following from the transitions

1. State: A = [2, 2, 2, 0, 0, 0] $\leftrightarrow$ B = (2, 1, 1, 1, 1, 0)

$$E_{dispersion} = \frac{8\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{8\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s \right)} + \frac{8\tilde{\delta}^2}{\sqrt{8 \left| \frac{\tilde{\beta}_N}{-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}_N^2}} \right|}}$$

2. State: A = (2, 1, 1, 1, 1, 0) $\leftrightarrow$ B = [2, 2, 2, 0, 0, 0]

$$E_{dispersion} = \frac{8\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{8\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s \right)} + \frac{8\tilde{\delta}^2}{\sqrt{8 \left| \frac{\tilde{\beta}_N}{-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}_N^2}} \right|}}$$

3. State: A = (2, 1, 2, 0, 1, 0) $\leftrightarrow$ B = (2, 1, 2, 0, 1, 0)

$$E_{dispersion} = \frac{4\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s + 2\tilde{\beta} - 2\tilde{\beta}_s \right)} + \frac{16\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{16\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s \right)} + \frac{8\tilde{\delta}^2}{\sqrt{8 \left| \frac{\tilde{\beta}_N}{-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}_N^2}} \right|}}$$

4. State: A = (2, 1, 2, 0, 1, 0) $\leftrightarrow$ B = (2, 1, 2, 1, 0, 0)

$$E_{dispersion} = \frac{16\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{8\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s \right)} + \frac{4\tilde{\delta}^2}{\sqrt{8 \left| \frac{\tilde{\beta}_N}{-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}_N^2}} \right|}}$$

5. State: A = (2, 1, 2, 0, 1, 0) $\leftrightarrow$ B = (2, 2, 1, 0, 1, 0)

$$E_{dispersion} = \frac{4\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s + 2\tilde{\beta} - 2\tilde{\beta}_s \right)} + \frac{8\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{12\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s \right)} + \frac{12\tilde{\delta}^2}{\sqrt{8 \left| \frac{\tilde{\beta}_N}{-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}_N^2}} \right|}}$$

6. State: $A = (2, 1, 2, 0, 1, 0) \leftrightarrow B = (2, 2, 1, 1, 0, 0)$

$$E_{dispersion} = \frac{8\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{4\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s \right)} + \frac{8\tilde{\delta}^2}{\sqrt{8 \left| \frac{\tilde{\beta}_N}{-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2}} \right|}}$$

7. State: $A = (2, 1, 2, 1, 0, 0) \leftrightarrow B = (2, 1, 2, 0, 1, 0)$

$$E_{dispersion} = \frac{16\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{8\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s \right)} + \frac{4\tilde{\delta}^2}{\sqrt{8 \left| \frac{\tilde{\beta}_N}{-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2}} \right|}}$$

8. State: $A = (2, 1, 2, 1, 0, 0) \leftrightarrow B = (2, 1, 2, 1, 0, 0)$

$$E_{dispersion} = \frac{16\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{16\tilde{\delta}^2}{\sqrt{8 \left| \frac{\tilde{\beta}_N}{-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}} \right|} + 1 \sqrt{8 \left| \frac{\tilde{\beta}_N}{-\tilde{\alpha}_s + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2}} \right|}}$$

9. State: $A = (2, 1, 2, 1, 0, 0) \leftrightarrow B = (2, 2, 1, 0, 1, 0)$

$$E_{dispersion} = \frac{8\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{8\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s \right)} + \frac{4\tilde{\delta}^2}{\sqrt{8 \left| \frac{\tilde{\beta}_N}{-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2}} \right|}}$$

10. State: $A = (2, 1, 2, 1, 0, 0) \leftrightarrow B = (2, 2, 1, 1, 0, 0)$

$$E_{dispersion} = \frac{8\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{20\tilde{\delta}^2}{\sqrt{8 \left| \frac{\tilde{\beta}_N}{-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}} \right|} + 1 \sqrt{8 \left| \frac{\tilde{\beta}_N}{-\tilde{\alpha}_s + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2}} \right|}}$$

11. State: $A = (2, 2, 1, 0, 1, 0) \leftrightarrow B = (2, 1, 2, 0, 1, 0)$

$$E_{dispersion} = \frac{4\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s + 2\tilde{\beta} - 2\tilde{\beta}_s \right)} + \frac{8\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{12\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s \right)} + \frac{12\tilde{\delta}^2}{\sqrt{8 \left| \frac{\tilde{\beta}_N}{-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2}} \right|}}$$

12. State: $A = (2, 2, 1, 0, 1, 0) \leftrightarrow B = (2, 1, 2, 1, 0, 0)$

$$E_{dispersion} = \frac{8\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{8\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s \right)} + 4\tilde{\delta}^2 \left(\frac{\tilde{\beta}_N}{\sqrt{8} \left| \frac{-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2}}{2} \right|} \right)$$

13. State: $A = (2, 2, 1, 0, 1, 0) \leftrightarrow B = (2, 2, 1, 0, 1, 0)$

$$E_{dispersion} = \frac{4\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s + 2\tilde{\beta} - 2\tilde{\beta}_s \right)} + \frac{4\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{8\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s \right)} + \frac{16\tilde{\delta}^2}{\sqrt{8}} \left(\frac{1}{\sqrt{8}} \right)$$

14. State: $A = (2, 2, 1, 0, 1, 0) \leftrightarrow B = (2, 2, 1, 1, 0, 0)$

$$E_{dispersion} = \frac{4\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{4\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s \right)} + 8\tilde{\delta}^2 \left(\frac{\tilde{\beta}_N}{\sqrt{8} \left| \frac{-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2}}{2} \right|} \right)$$

15. State: $A = (2, 2, 1, 1, 0, 0) \leftrightarrow B = (2, 1, 2, 0, 1, 0)$

$$E_{dispersion} = \frac{8\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{4\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s \right)} + \frac{8\tilde{\delta}^2}{\sqrt{8} \left| \frac{\tilde{\beta}_N}{-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}_N^2}} \right|}$$

16. State: $A = (2, 2, 1, 1, 0, 0) \leftrightarrow B = (2, 1, 2, 1, 0, 0)$

$$E_{dispersion} = \frac{8\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{20\tilde{\delta}^2 \left(\sqrt{8 \left| \frac{\tilde{\beta}_N}{-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}} \right|^2} + 1 \sqrt{8 \left| \frac{\tilde{\beta}_N}{-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}} \right|^2} \right)}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)}$$

17. State: $A = (2, 2, 1, 1, 0, 0) \leftrightarrow B = (2, 2, 1, 0, 1, 0)$

$$E_{dispersion} = \frac{4\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{4\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s \right)} + 8\tilde{\delta}^2 \left(\sqrt{8 \left| \frac{\tilde{\beta}_N}{-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}_N^2}} \right|} \right)$$

18. State: A = (2, 2, 1, 1, 0, 0) ↔ B = (2, 2, 1, 1, 0, 0)

$$E_{dispersion} = \frac{4\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)} + \frac{16\tilde{\delta}^2 \left(\sqrt{8 \left| \frac{\tilde{\beta}_N}{-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}} \right|^2} + 1 \sqrt{8 \left| \frac{\tilde{\beta}_N}{-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}} \right|^2} + 1 \right)}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)}$$

Ground-State:

A = [2, 2, 2, 0, 0, 0] ↔ B = [2, 2, 2, 0, 0, 0]

$$E_{dispersion} = \frac{16 \left(\sqrt{8 \left| \frac{\tilde{\beta}_N}{-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}} \right|^2} + 1 \sqrt{8 \left| \frac{\tilde{\beta}_N}{-\tilde{\alpha} + \tilde{\alpha}_N + \tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_N - 2\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_N^2 + 2\tilde{\alpha}_N\tilde{\beta} + \tilde{\beta}^2 + 8\tilde{\beta}_N^2}} \right|^2} + 1 \right)}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s \right)}$$